

Mn-Based Molecular Catalysts for the Electrocatalytic Disproportionation of CO₂ into CO and CO₃²⁻

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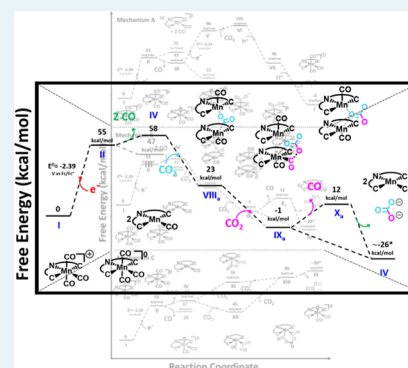
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ABSTRACT: This work studies the electrocatalytic disproportionation of CO₂ into CO and CO₃²⁻ mediated by a [Mn-2,6-bis(1-(alkyl)imidazol-2-ylidene)pyridine]Br pincer complex. We identify three mechanistic scenarios involving one or two catalytically active metal centers in the activation of carbon dioxide and use density functional theory to map out the energy landscape for each of the mechanistic steps. Experimentally determined second-order kinetics in CO₂ consumption, the formation of CO and CO₃²⁻ together with an observed order in catalyst of 0.5 determined by cyclic voltammetry using the Burš normalized timescale method, suggest a turnover-limiting CO₃²⁻ loss through the scission of a bimetallic species. Faradaic efficiencies for the reduction of CO₂ to CO of 86 ± 4% are observed. Given the reaction mass balance, we estimate a Faradaic efficiency from the conversion of CO₂ to CO₃²⁻ of 93 ± 4%. Our combined experimental–theoretical approach suggests that two sequential CO₂ insertions are followed by a rearrangement to produce carbon monoxide and carbonate from two molecules of CO₂ at singly reduced catalyst molecules. The disproportionation reaction reported herein combines the p sequestration and provides a new chemical avenue for the conversion of carbon dioxide.

KEYWORDS: electrocatalysis, solar fuels, carbon dioxide conversion, manganese catalysis



■ INTRODUCTION

High levels of CO₂ in the atmosphere are leading to severe environmental challenges including global climate change and ocean acidification.^{1,2} CO₂ is an abundant C₁ source that has great potential for the synthesis of value-added chemicals. Thus, it has been the focus of research driven by current societal and environmental needs for addressing rising atmospheric CO₂ levels and producing chemical fuels or precursors from abundant sources. Despite considerable research focused on the catalytic reduction of CO₂, its linear geometry and stability continue to pose challenges to potentially practical approaches for CO₂ reduction. Electrocatalysis has been shown to be one potentially viable option for the conversion of CO₂ to reduced products.^{3,4}

Both homogeneous and heterogeneous electrocatalysts have been studied for CO₂ conversion. Although heterogeneous catalysts tend to have high activities, they typically have low selectivity, often simultaneously producing a variety of products such as H₂, methane, ethanol, acetate, and propanol.⁵⁻⁸ Homogeneous catalysts are a possible alternative that benefit from the molecular tunability of both the metal center and the ligand framework. Nevertheless, they suffer from low current densities, which limit the practical applicability of current state-of-the-art systems. Therefore, researchers have targeted discovering catalysts with increased

current density, in addition to higher selectivity and long-term stability.^{3,5}

This work studies the mechanism of CO₂ reduction catalyzed by [MnCNC^{Bn}(CO)₃]Br (**1**) shown in Figure 1.⁹ This complex is of particular interest due to its significant catalytic rates and ability to reduce carbon dioxide in the absence of a proton source.

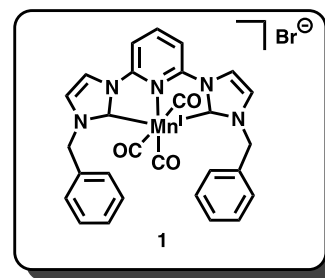


Figure 1. Structure of the octahedral Mn^I complexes of interest, [MnCNC^{Bn}(CO)₃]Br (**1**).

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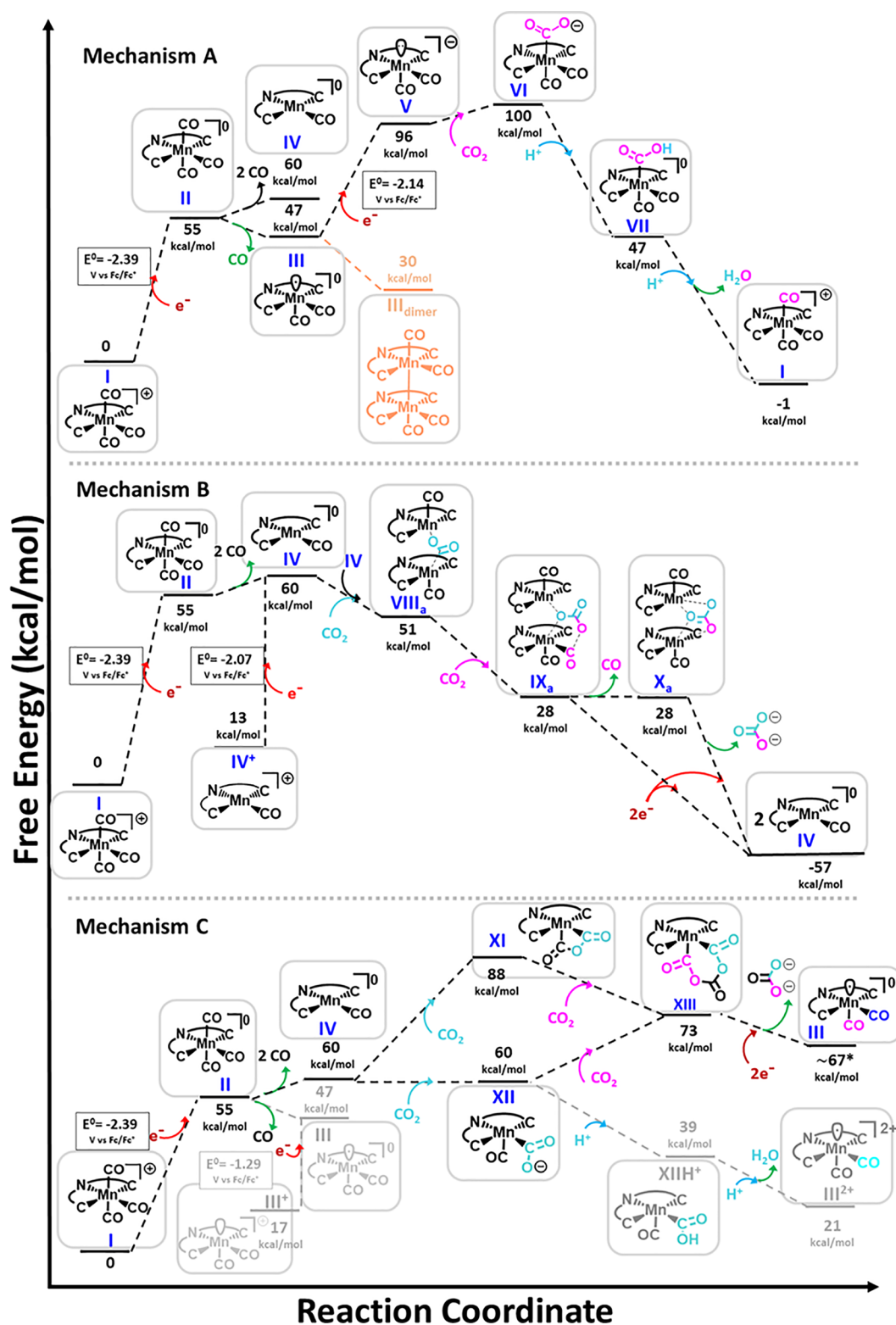


Figure 2. Reaction coordinate diagram of the three CO₂RR mechanisms investigated. Mechanism A is a proton-mediated reduction, which produces CO and H₂O. Mechanism B describes a binuclear proton-less reduction pathway, and mechanism C operates within a mononuclear proton-less scenario. Both mechanisms B and C are consistent with the experimental observation of second-order kinetics with respect to CO₂ consumption and explain the formation of CO and CO₃²⁻. *ΔG_{CO₃release} has been calculated as the sum of three ΔG's: ΔG_{dissociation} + ΔG_{stabilization} + ΔG_{reduction}, where ΔG_{dissociation} is the free energy to dissociate the LMn–CO₃ complex into its respective Mn^{+1/+2} intermediate(s) and solvated CO₃²⁻, ΔG_{stabilization} is the free energy to stabilize the coulombic interactions between CO₃²⁻ and 2N(*n*-Bu)₄⁺, and ΔG_{reduction} is the electrochemical free energy added to Mn^{+1/+2} intermediates after dissociation. A legend of all calculated intermediates is available in Section S7.

Mechanisms involving several group VII catalysts, such as $[M(\text{bpy})(\text{CO})_3]\text{X}$ and $[M(\text{tBu-bpy})(\text{CO})_3]\text{X}$ (where $M = \text{Re}$, Mn and $\text{X} = \text{Cl}$, Br), have been studied.^{10–13} In a report by Richeson and co-workers, an instance of proton-less carbon dioxide reduction was described as catalyzed by a Mn tricarbonyl pincer compound: $[\text{Mn}(\kappa^3\text{-2,6-(Ph}_2\text{PNMe)}_2(\text{NC}_5\text{H}_3))(\text{CO})_3]\text{Br}$. The mechanism for this reaction, however, has not yet been reported.¹⁴ Franco et al. proposed a proton-less disproportionation mechanism for the compound $[\text{Mn}(\text{CO})_3(\text{bis-}^{\text{Me}}\text{NHC})\text{Br}]$, which was suggested to proceed through mononuclear intermediates.¹⁵ The absence of protons is important for several reasons. First, reactions requiring a proton source must be continuously supplied with protons at the correct concentration to maintain maximum efficiency. Second, catalysts that require protons often produce H_2 as a side product, thus lowering their Faradaic efficiency.^{16–18} In contrast, a catalyst that operates in the absence of protons avoids the issue of hydrogen selectivity of the reduction process. Additionally, the absence of protons avoids slow proton transfers from limiting the rate of CO_2 conversion.⁴ Additionally, proton-less reduction mechanisms can form oxalate and are of interest for carbon–carbon coupling reactions via electrocatalytic reduction to produce liquid fuels.¹⁸

■ RESULTS AND DISCUSSION

With these previously proposed mechanisms in mind, we performed an experimental and theoretical study on the CO_2 reduction reaction (CO_2RR) mechanism catalyzed by complex **1**. We examined three plausible CO_2RR mechanisms by **1** $[\text{Mn}(\text{CNC}^{\text{Bn}})(\text{CO})_3]\text{Br}$;⁹ mechanism A involves a proton-mediated reduction (Figure 2, top), mechanism B involves the activation of CO_2 by two Mn metal centers in the absence of protons (Figure 2, middle), and mechanism C describes a proton-less mononuclear pathway (Figure 2, bottom). Figure 2 shows the reaction coordinate diagrams for these mechanistic scenarios.

We proceeded with the assumption that the energy differences between methyl and benzyl substituents are negligible due to the large distances from the active Mn center. Therefore, our electronic structure calculations employ a model of $[\text{Mn}(\text{CNC}^{\text{Me}})(\text{CO})_3]\text{Br}$ instead of $[\text{Mn}(\text{CNC}^{\text{Bn}})(\text{CO})_3]\text{Br}$.

Mechanism A. Density functional theory (DFT) calculations reveal the relative energies of the proposed intermediates **II**–**VII** of a proton-mediated cycle (Figure 2, top panel). The starting point of the catalytic cycle is $[\text{Mn}(\text{CNC}^{\text{Me}})(\text{CO})_3]^+$, denoted as **I**. First, **I** is electrochemically reduced to **II** upon the application of -2.39 V vs Fc/Fc^+ (or -2.23 V vs Fc/Fc^+ ; see SI). This is equivalent to an increase in the free energy of 55 kcal mol^{-1} and is likely the potential-determining step. Experiments corroborate this first $1e^-$ reduction, with an observed potential of -2.1 V vs Fc/Fc^+ for $[\text{Mn}(\text{CNC}^{\text{Me}})(\text{CO})_3]^+$.¹⁶ The number of electrons exchanged in this first reduction event was determined using normal pulse voltammetry and diffusion ordered spectroscopy (DOSY NMR) by the method reported by Donadt and co-workers.^{9,19}

DFT calculations of the species indicate that upon a $1e^-$ reduction of **I**, the electron is localized on the ligand, as indicated by spin density population calculations (Figure S14). This reduction event is followed by a metal-to-ligand charge transfer (MLCT), which results in the formation of the

trigonal bipyramidal species **III** and the subsequent release of a CO molecule.

The potential required to reduce **III** to **V** ($E_{\text{III}/\text{V}}^0 = -2.14$ V vs Fc/Fc^+) is 0.25 V less negative than the applied potential required to reduce **I** and thus this reduction is favorable. We hypothesize that species **III** may also be involved in a dimerization reaction, as its Mn center now has a singly occupied radical-like orbital. However, dimerization events are less likely to occur compared to further electrochemical reduction of **III** and subsequent formation of intermediate **V**. This is because the time scales of electron-transfer processes in electrochemical reduction are significantly shorter than those of nuclear motion associated with dimerization. Nevertheless, our calculations predict that the oxidation of the dimer may occur at approximately -1.6 V vs Fc/Fc^+ , suggesting that the dimer species may exist in small amounts, as observed in Figure S1 of the Supporting Information (SI).

The second electrochemical reduction step of mechanism A generates species **V**, which, in turn, proves sufficiently nucleophilic to activate CO_2 via an attack to form adduct **VI**. We compute the activation of CO_2 by the Mn center to be endergonic by 4 kcal/mol. In addition, **VI** possesses a similar geometry to **I**; however, one of its axial CO ligands is replaced by CO_2^- . CO_2 is more likely to approach from an axial position as opposed to an equatorial attack, which is impeded by CO ligands that are unable to react with CO_2 . Equatorial CO_2 binding consequently requires an additional 13 kcal/mol relative to axial attack.

After the catalyst successfully activates CO_2 , the negatively charged CO_2 fragment of **VI** can be protonated. Our calculations indicate that the proton transfer to produce a carboxylate ligand is highly favorable and can be accomplished with weak acids ($\text{pK}_a < \sim 29$, MeCN). A second protonation then occurs, leading to a dehydration reaction. Subsequently, water is formed and released, leaving a CO ligand and the return of the catalyst to its resting state **I**.

While mechanism A is plausible, the observed turnover at the $1e^-$ step and catalysis in the absence of added protons (Figure S1) is puzzling. With no evidence for the overall mass balance of the reaction or the kinetic orders in reaction components, we considered further experiments to probe whether mechanism A was in effect. We next examined whether any CO was released upon the first electron reduction. After delivering a $1e^-$ equivalent under an argon atmosphere, we produced 2 equivalents of CO, suggesting the formation of a Mn species with a single remaining carbonyl group, most likely **IV**, rather than species **III**.⁹ We then proceeded to determine the kinetic orders in CO_2 for this reduction using cyclic voltammetry of **I** under Ar and CO_2 gas mixtures of varying $\text{CO}_2\%$ volume fractions from 10% to 100%. The results are shown in Figure 3. Although at higher concentrations of carbon dioxide, the rate of consumption does not change, this is likely due to the gradual saturation of carbon dioxide at high CO_2 fractions. At low concentrations, the reaction is shown to be second order with respect to CO_2 .

Using the normalized timescale method,²⁰ we proceeded to determine the order in catalyst for this reaction. Given adequate Faradaic efficiencies, we assume that the charge passed over time in the electrochemical experiment can be directly used to determine relative rates between measurements with varying catalytic conditions. Cyclic voltammograms were collected at concentrations of 0.24, 0.5, 0.75, 1, and 1.5 mM catalysts at CO_2 saturation in acetonitrile. Plots of charge

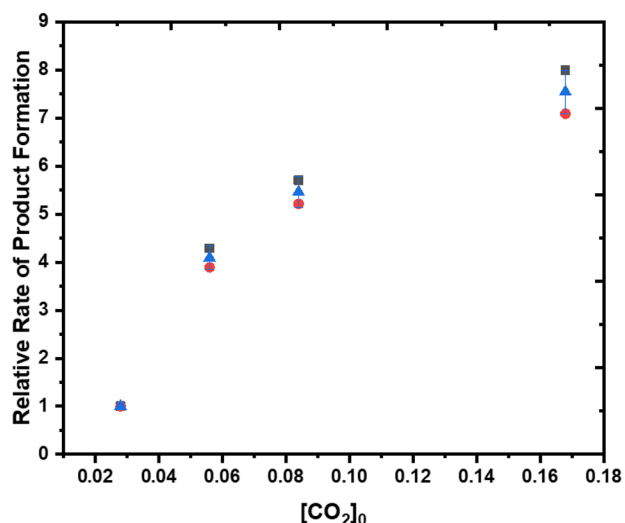


Figure 3. Graphical representation of a series of cyclic voltammograms using 2 mM of **1** at different $[\text{CO}_2]$. At lower concentrations of CO_2 , doubling $[\text{CO}_2]$ increased the relative rate by a factor of 4, indicative of a second-order reaction.

passed vs time were generated for each of the runs from the catalytic onset, and the x-axis was multiplied by the concentration of the catalyst raised to the power of the kinetic orders of 0.5, 1, and 2 to determine the overlay that identifies the reaction order. An order of 0.5 was determined by this method through the overlay of charge vs time $\times [\text{cat}]^{\text{order}}$ as shown in Figure 4. This order suggests that the rate-determining step is likely the breakdown of a bimetallic species, likely the release of a carbonate molecule sandwiched in between the MnCNC cores.

Although reaction mechanism A is a plausible scenario, it does not explain (a) turnover at a $1e^-$ step, (b) the second-order CO_2 dependence, (c) the production of CO in the absence of a proton source, (d) the formation of CO_3^{2-} product, and (e) the observed 0.5 order in catalyst.

To quantify the carbonate signals observed by FTIR (Figures S3–S5), derivatization was performed akin to the literature precedent,²¹ and the samples were analyzed by gas chromatography with electron impact detection (GC-EI). Standard solutions of CO_3^{2-} for calibration were prepared using sodium carbonate in water. For quantification, analyte solutions from electrolysis were dried via rotary evaporation followed by a high vacuum and then dissolved in 20 mL of HPLC-grade water to maintain the original concentration following electrolysis. The analysis of the data obtained led to the determination of a Faradaic efficiency of CO_2 to CO_3^{2-} of $93 \pm 4\%$.

With this information in hand, we proceeded to consider the work of Fujita and co-workers, in which an alternate mechanism is proposed based on an intriguing stoichiometric and computational study of photochemical carbon dioxide conversion at a $\text{Re}^{0/0}$ dimer that did not involve protons.²² In this instance, the reduction of CO_2 formed CO and CO_3^{2-} .^{10,11,13} In contrast, a related $\text{Mn}(\text{bpy})(\text{CO})_3\text{Br}$ complex did not reduce CO_2 in the absence of protons¹³ and readily formed a $\text{Mn}^{0/0}$ dimer, which had been shown to be competent toward CO_2 reactivity.^{12,13,23,24} Carbonate has also been observed in the electrochemical reduction of carbon dioxide by cyclic(alkyl)(amino) carbenes in both anhydrous THF and MeCN²⁵ and in catalyses by Re and Mn complexes of (α -diimine) $\text{M}(\text{CO})_3\text{Br}$.²⁶

Mechanism B. Given the experimental data, we revised our mechanistic hypothesis to mechanism B (Figure 2, middle).

In relation to mechanism B in Figure 2 (middle) and Figure S5, experimental observations indicate the loss of two CO molecules after the first $1e^-$ reduction of **I**,⁹ which we compute to be endergonic by ~ 5 kcal/mol; following the $1e^-$ reduction of complex **I**, the square planar complex **IV** is formed. Our calculations predict that **IV** is more reactive than **V** (mechanism A) toward CO_2 and exhibits a unique antiferromagnetic behavior as discussed in the SI (Figure S24). Upon generation of **IV**, CO_2 is easily activated when sandwiched between two **IV** intermediates to form complex **VIII_a**, a complexation we compute to be 14 kcal/mol exergonic. Because trimolecular reactions are not probable, the most likely pathway involves a stepwise mechanism of a thermoneutral insertion of CO_2 to **IV** to form intermediate **XII** (Figure 2, mechanism C), which is then stabilized by another **IV** complex, where Mn acts as a Lewis acid, to enable Mn–C–O–Mn connectivity in complex **VIII_a**. The first CO_2 insertion can occur in various orientations. The SI provides additional details about other possible insertion/reaction paths. A second CO_2 insertion is also favorable as suggested by the computed negative ΔG_{rxn} for the transformation of **VIII** into **IX_a**. Both CO_2 insertions are predicted to be barrier-less reactions as evidenced by the lack of saddle points in scans along the reaction coordinates performed to locate a transition state (Figures S26–S34).^{27,28} The two inserted CO_2 molecules effectively disproportionate into a carbonate anion that is sandwiched between two planar **IV** molecules and a CO that is bound to a remaining Mn complex fragment (Figure 2 middle and Figures S18 and S19).

The carbonate release step can be represented by the following chemical equation

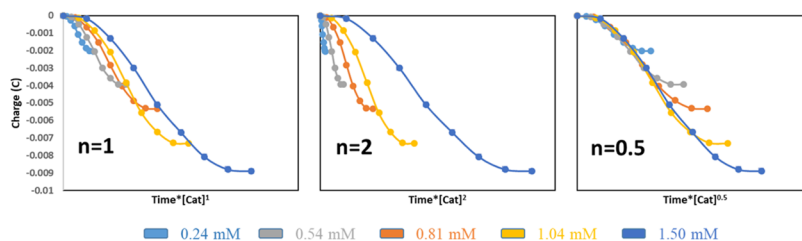
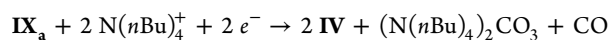


Figure 4. Graphical representation of $\text{time} \times [\text{Cat}]^{\text{kinetic order}}$ for the determination of order in catalyst through visual overlay using the Burés normalized time scale method.²⁰

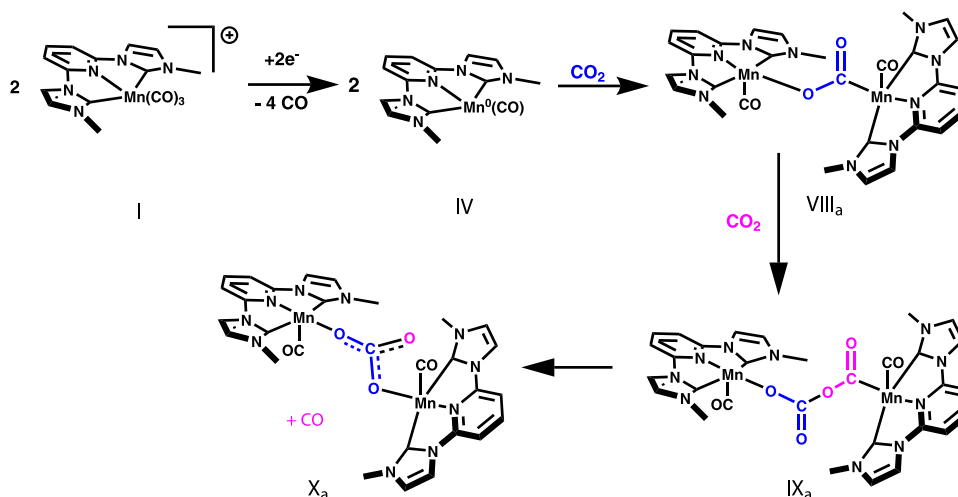


Figure 5. Mechanism B: proposed reduction mechanism beginning with $[\text{Mn}^{\text{I}}\text{CNC}^{\text{Me}}(\text{CO})_3]^+$ (**I**). Two complexes **I** are reduced by a total of $2e^-$ ($1e^-$ each) to produce 4 CO equivalents and form two complexes **IV**, which then react with CO_2 to form **VIII_a**. A second CO_2 then inserts into complex **IX_a** followed by a rearrangement to produce the bimetallic carbonate complex **X_a** and CO. The overall reaction is a disproportionation that converts two CO_2 molecules into CO_3^{2-} and CO.

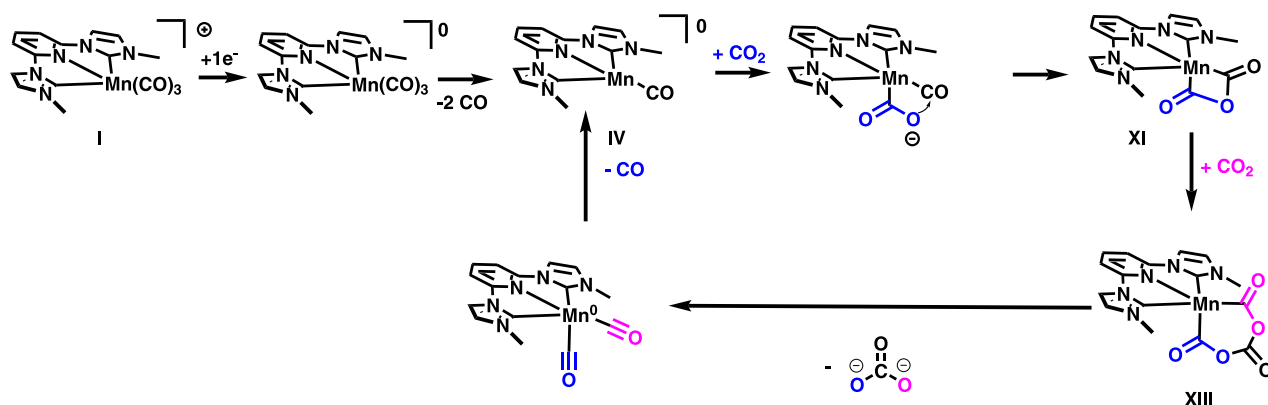


Figure 6. Mechanism C, a mononuclear CO_2 reduction pathway. Relative energies for intermediates are provided in Figure 2, bottom panel.

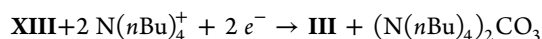
and is divided into three elementary steps: (a) the dissociation of **IX_a** (or **X_a**) into CO_3^{2-} and electron-deficient variants of **IV**, (b) the stabilization of CO_3^{2-} by $\text{N}(n\text{-Bu})_4^+$ cations, which are present as NBu_4PF_6 electrolyte in solution, and (c) the electrochemical regeneration of **IV**. The dissociation step that forms the charged species is endergonic due to the unfavorable dissociation of bonds and is the only uphill step after the formation of **IV**, indicating that it is most likely the rate-determining step of mechanism B. Because this step succeeds two CO_2 insertions, it is consistent with the experimental observations, which suggest that the rate of CO production is second order in CO_2 concentration and an order in catalyst of 0.5. Dissociation is endergonic by 34 kcal/mol, but this is partially alleviated by the stabilizing coulombic interactions between CO_3^{2-} and two neighboring $\text{N}(n\text{-Bu})_4^+$ cations, which is favorable by ~ 24 kcal/mol, bringing the thermodynamics of this chemical process to overall +10 kcal/mol endergonic. The electrochemical process of reducing two **IV⁺** complexes to **IV** supplies sufficient energy in the form of two electrons, each at $E^0 = -2.07$ V vs Fc/Fc^+ , to drive the unfavorable reaction forward. We calculated the total free energy of the three steps as the sum of (a) $\Delta G_{\text{dissociation}} = 34$ kcal/mol, (b) $\Delta G_{\text{stabilization}} = -24$ kcal/mol, and (c) $\Delta G_{\text{reduction}} = -(2 \times 48)$ kcal/mol, which results in a $\Delta G_{\text{CO}_3\text{release}}$ of ~ 85 kcal/mol downhill,

indicating that the overall process of carbonate release is favorable. This analysis is nevertheless limited by a few assumptions. First, the dissociation must occur near the electrode such that the electrons are rapidly supplied to **IV⁺** to regenerate **IV**. Second, the carbonate must be immediately stabilized by $\text{N}(n\text{-Bu})_4^+$. We believe this occurs because the positively charged $\text{N}(n\text{-Bu})_4^+$ ions are attracted to the negatively charged cathode by coulombic forces but do not reside too close to the surface due to their large size.

Mechanism C. While mechanism B is corroborated by experimental observations, an alternative mechanistic pathway that proceeds via a mononuclear mechanism (mechanism C) is also plausible and shown in Figure 2, bottom panel, and Figure 6. In this reaction sequence, the reduction of the starting compound **I** is followed by the loss of two CO molecules to form **IV**. Intermediate **IV** binds a CO_2 molecule that could form a metalanhydride **XI**, a process that is endergonic by 28 kcal/mol from the ground state of **IV**. Thus, **XI** is a highly unlikely intermediate in this system. The thermoneutral formation of compound **XII** from **IV** hints that the CO_2 sandwiching step in mechanism B could, in fact, be stepwise, where intermediate **XII** is first formed followed by an approaching **IV** to form **VIII_a**. The insertion of a second CO_2 into a Mn–C bond of **XII** to form **XIII** is endergonic by

13 kcal/mol, although the subsequent carbonate release and CO production are a 7 kcal/mol downhill. Consequently, this mechanism is highly unlikely relative to mechanisms A and B due to the large positive ΔG_{rxn} associated with forming intermediate XIII. Protonation of oxygen atoms in complex XII is possible with strong acids as the protonated form XIIH⁺ has a pK_a value of ~11 in MeCN, allowing a different proton-mediated pathway to form CO via the IV intermediate. However, this pathway yields III²⁺ as a product, which can be easily reduced back to III but requires either a penalty of 13 kcal/mol to regenerate IV, or induce another electrochemical reduction to V to close the catalytic cycle. This suggests that even though this pathway is favorable, mechanism A remains the more likely proton-mediated reduction path.

The carbonate release is computed similarly to that of mechanism B using the following equation



One main difference is that in mechanism B, X_a dissociates into carbonate and two IV⁺, whereas in mechanism C, XIII dissociates into carbonate and one III²⁺ complex. Unlike mechanism B, the carbonate in mechanism C is formed by a combination of an original CO ligand and oxygen atoms from both inserted CO₂ molecules. Both dissociation reactions are similarly highly unfavorable: $\Delta G_{\text{diss, Mech2}} = 34$ kcal/mol versus $\Delta G_{\text{diss, Mech3}} = 46$ kcal/mol. However, the electrochemical reductions to reduce two IV⁺ complexes to two IV complexes add significantly more energy to the system compared to reducing III²⁺ twice to III. Although in both cases, two electrons are added, the first reduction of III²⁺ to III⁺ occurs at near 0 V vs Fc/Fc⁺ and thus does not completely harness the energy of the electron at the applied potential. The second reduction of III⁺ to III occurs at −1.29 V vs Fc/Fc⁺. On the other hand, the reduction of IV⁺ to IV occurs at −2.07 V vs Fc/Fc⁺, indicating that more than twice the energy is added electrochemically in mechanism B than in mechanism C and explaining why the former's carbonate release is very favorable, whereas the latter's is less favorable and is thus less likely to occur. In addition, mechanism C is inconsistent with an observed half-order in catalyst.

CONCLUSIONS

We have examined a set of three mechanistic proposals to explain the reduction of carbon dioxide as catalyzed by complex 1. The first CO₂RR mechanism we studied is a proton-mediated reduction similar to previously proposed mechanisms; however, the pathway does not explain the observed turnover after a single-electron reduction and in the absence of added protons. Mechanisms B and C account for: catalytic scenarios in the absence of protons, the 1e[−] reduction per Mn center, the observation that the reaction is second order in CO₂ consumption, and the formation of carbonate with a 93% Faradaic efficiency. Of these two possibilities, our calculations predict that mechanism B is more energetically favorable. Mechanism B is consistent with a 0.5 order in catalyst, thus indicating that the rate-determining step is likely the loss of carbonate from a bimetallic species. Mechanism B is a combination of catalytic conversion and sequestration, both of which are highly desirable processes being developed potentially to counteract rising levels of CO₂.

Although device engineering for CO₂ reduction remains an active area of research, proton conductivity in membranes that

separate anode and cathode chambers remains a major concern. While CO₂ solubility in ionic aqueous media is a limiting factor in the development of scalable technologies, organic solvents avoid this limitation. When considering device operation under nonaqueous conditions, catalytic systems that enable proton-less operation alleviate the need for proton transporting membranes, therefore paving the way toward the engineering of new CO₂ conversion processes. Additionally, second-order reactions have been harnessed for the formation of C₂ products such as ethanol and ethylene. Therefore, the mechanistic step in which a second carbon dioxide is inserted into a Mn–C bond raises the intriguing possibility of tailoring these systems toward potential C–C bond formation chemistry in the near term.^{29,30}

METHODS

General Methods. Except where noted, reactants were used as received from commercial sources without further purification. Synthesis of the ligands and the metal complex were performed as previously reported.⁹ Tetrabutylammonium hexafluorophosphate (TBAH) was recrystallized from ethanol and then dried under a high vacuum for 24 h before use. Metal complexes were stored in low-light conditions to prevent light exposure and degradation. All NMR spectra were gathered on a Bruker Avance-III 300 MHz NMR spectrometer at room temperature with chemical shifts reported with respect to internal proton or carbon solvent for ¹H and ¹³C NMR spectra, respectively.

Electrochemical Experiments. A standard three-electrode setup with a glassy carbon working electrode (3 mm, Bioanalytical Systems), a platinum wire counter electrode, and a silver wire single-junction pseudoreference was used for cyclic voltammetry (CV) and normal pulse voltammetry (NPV). For electrolysis experiments, a carbon cloth counter electrode was used in place of the platinum wire. Ferrocene was used as an external standard. All experiments were performed with the exclusion of air using an argon purge. In the case of NPVs, concentrations were adjusted for the evaporated solvent to calculate the number of electrons accurately.^{9,19} All data workup was done via OriginPro v9.4.0.220 software.

Carbonate Quantification. The derivatization method for the quantification of carbonate was performed akin to literature precedent.²¹ Standard solutions of CO₃^{2−} for calibration were prepared using sodium carbonate in water. For quantification, analyte solutions from electrolysis were dried via rotary evaporation followed by high vacuum drying. The residue was then dissolved in 20 mL of HPLC-grade water to the original concentration following electrolysis. For derivatization, 100 μL of solution was mixed with 800 μL of deuterated acetone (CD₃COCD₃) and 20 μL of 2,3,4,5,6-pentafluorobenzyl bromide. This mixture was then heated to 50 °C for 60 min in an aluminum heating block. Acetone was then removed from the solution under a stream of argon, and toluene was used to extract the reaction products. A portion of the toluene aliquot was then diluted 1:50 for GC–MS injection and analyzed on a Thermo Scientific GC–EI.

Computational Methods. DFT calculations were performed using the Gaussian 16^{31,32} computational chemistry package based on the MN15 density functional.^{31,32} The Stuttgart/Dresden SDD basis set/pseudopotentials³³ were used to describe Mn atoms, and the 6-31+G(d,p) double-zeta basis set³⁴ was used to describe all other atoms.

Vibrational force constants at this level of theory were calculated to (a) validate that all geometries have only positive vibrational frequencies and (b) to compute entropies, zero-point energies, and thermal corrections for the reported free energies at 298 K. The CPCM implicit polarized continuum solvation model was used to simulate the solvation effects of acetonitrile.^{34,35} Refer to the **Methods** section and the **SI** for further details on free-energy calculation methods.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.9b04773>.

Synthetic and experimental procedures, spectra, and additional plots (PDF)

Details related to the calculations (ZIP) (ZIP)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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